

Deploying Web App on ECS - Step 8: Enable Service Auto Scaling (Scheduled Scaling)

To Begin with the Lab

Summary of the Lab

In this step, **Amazon ECS Service Auto Scaling** was configured using **scheduled scaling** so the application could increase and decrease task count at specific times. The lab first set minimum and maximum task limits for the service, then created a scheduled scale-out action that increased the desired count from 2 to 4 tasks. After that, a scheduled scale-in action was created so the service could return from 4 tasks back to 2 tasks later. The running tasks were verified through the service details and the **Application Load Balancer** target group, which automatically registered the new tasks. This showed that scheduled scaling can manage predictable traffic changes without needing manual intervention or load-based alarms.

Understanding Scheduled Scaling in Amazon ECS

Scheduled scaling in **Amazon ECS** lets you increase or decrease the number of running tasks at predefined times. It exists for workloads that follow a predictable pattern, such as business hours, weekends, or nightly low-traffic periods. Instead of watching CPU or memory like target tracking or step scaling, scheduled scaling uses time-based actions to scale out and scale in automatically. For example, you can increase tasks before peak usage starts and reduce them again when traffic becomes low, which keeps the app available while saving cost.

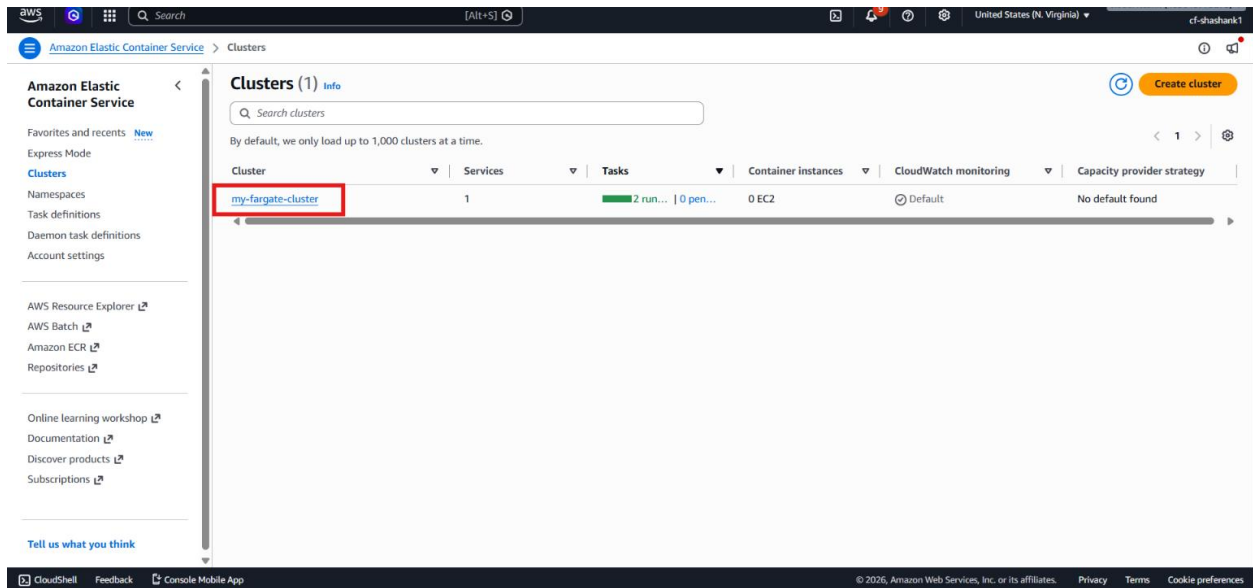
Example:

A PHP web app is running with 2 tasks during normal hours. At 22:55, a scheduled action increases the service to 4 tasks, and later at 23:00 another scheduled action reduces it back to 2 tasks. The **Application Load Balancer** automatically keeps routing traffic to the healthy tasks during both changes.

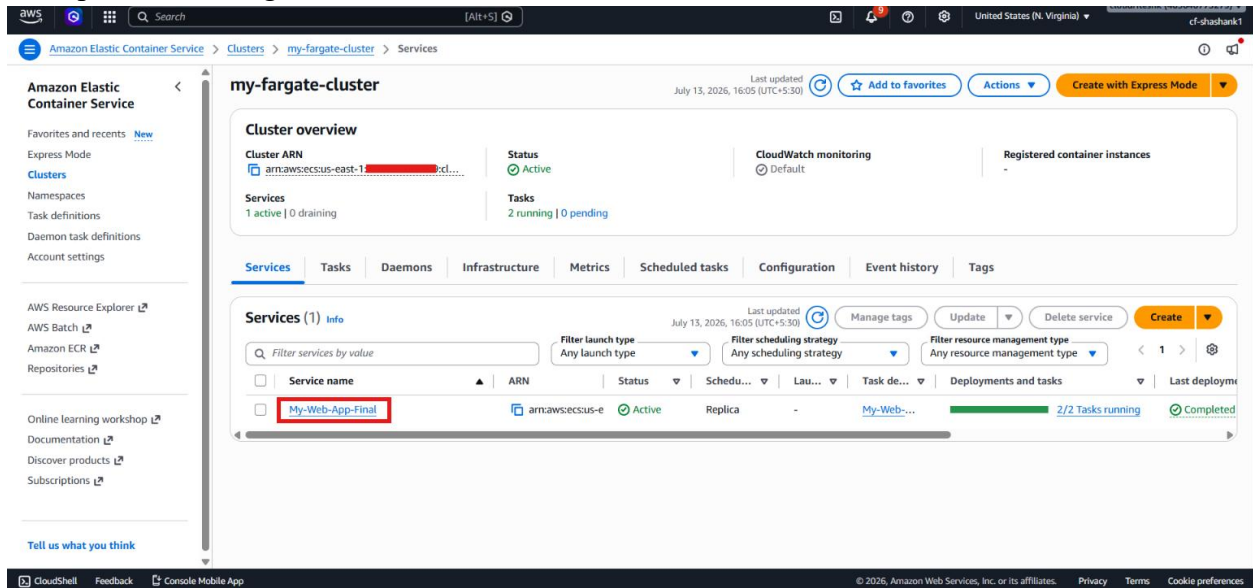
Lab Steps

Step 1 – Open ECS service auto scaling

- Sign in to the **AWS Management Console**.
- Open **Amazon ECS**.
- Open the Fargate cluster.



- Open the running service.



- Go to Service auto scaling.

Amazon Elastic Container Service

My-Web-App-Final

Status: Active

Tasks (2 Desired): 2 running | 0 pending

Task definition: revision My-Web-App:1

Deployment status: Success

Service auto scaling

How to use service auto scaling

Task count

Number of tasks to run for My-Web-App-Final: Not set

Set the number of tasks

Step 2 – Set the desired task limits

- Click Set the number of tasks.

Amazon Elastic Container Service

My-Web-App-Final

Status: Active

Tasks (2 Desired): 2 running | 0 pending

Task definition: revision My-Web-App:1

Deployment status: Success

Service auto scaling

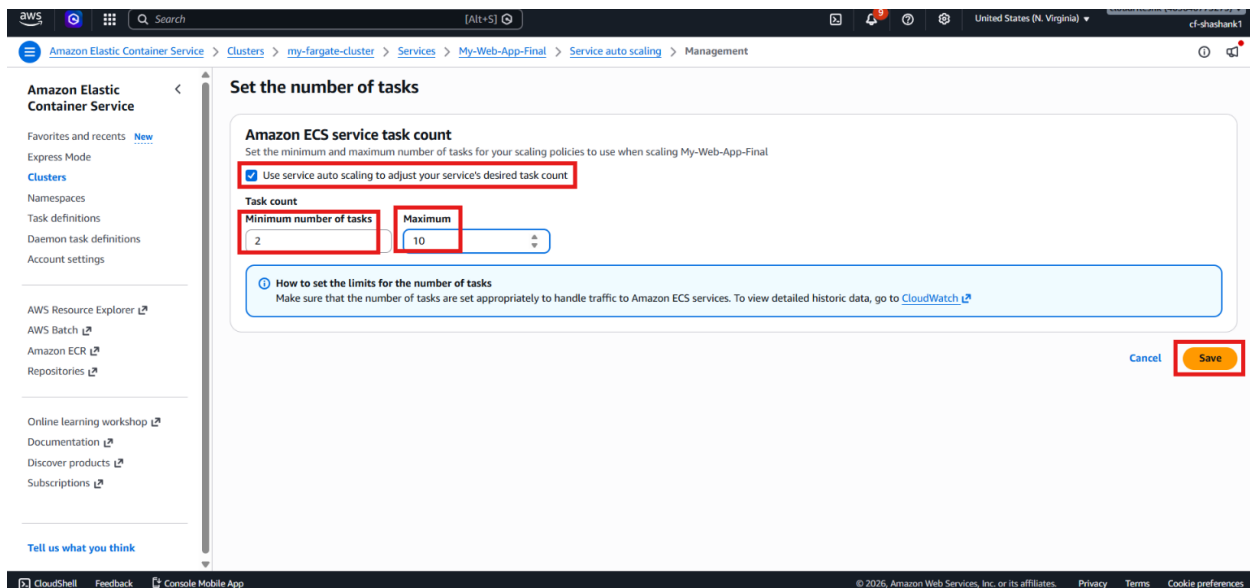
How to use service auto scaling

Task count

Number of tasks to run for My-Web-App-Final: Not set

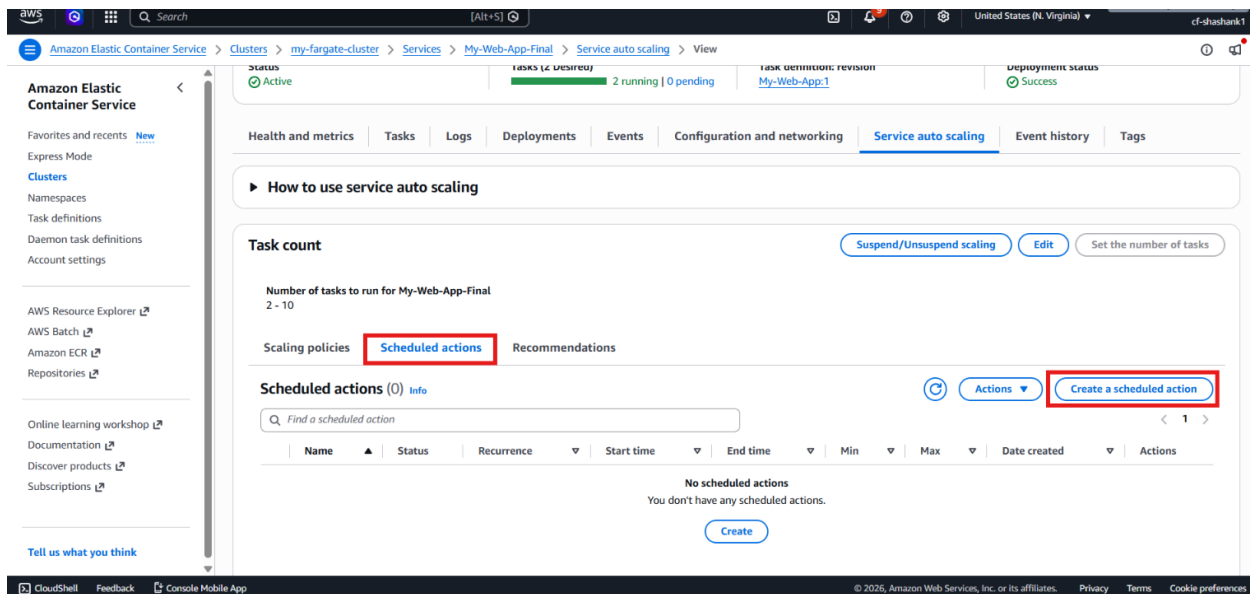
Set the number of tasks

- Set the minimum task count to 2.
- Set the maximum task count to 10.
- Save the task limits.



Step 3 – Create the scale-out scheduled action

- Click Create schedule action.



- Enter the scale-out policy name: SC-Weekend-Scale-Out
- Set the time zone to Asia/Calcutta.
- Choose today's date.
- Set the trigger time for the scale-out action.

Create scheduled action Info

Scheduled actions add or remove capacity automatically at preset times or intervals. Amazon Elastic Container Service autoscaling is using the following Service-linked role: [arn:aws:iam::463646775279:role/aws-service-role/ecs.application-autoscaling.amazonaws.com/AWSApplicationAutoScaling_ECSService](#)

Action details

Action name
SC-Weekend-Scale-Out

Schedule
Specify a schedule for your capacity changes.

Time zone
Asia/Calcutta

Current time in selected time zone is 07/13/2026, 16:11 GMT+5:30

Start time (GMT+5:30)
Schedule a specific date and time for the first scheduled action to run

Date
2026/07/13
Format: YYYY/MM/DD

Time
16:20:00
Format: hh:mm:ss

Recurrence
Once

Capacity adjustments
If the current capacity is below the minimum capacity at the scheduled time, capacity will scale out to the minimum capacity limit. If the current capacity is above the maximum capacity, it will scale in to the maximum capacity.

- Set the desired count to increase from 2 to 4.
- Save the scheduled scale-out action.

Create scheduled action Info

Scheduled actions add or remove capacity automatically at preset times or intervals. Amazon Elastic Container Service autoscaling is using the following Service-linked role: [arn:aws:iam::463646775279:role/aws-service-role/ecs.application-autoscaling.amazonaws.com/AWSApplicationAutoScaling_ECSService](#)

Action details

Action name
SC-Weekend-Scale-Out

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Specify a schedule for your capacity changes.

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Asia/Calcutta

Current time in selected time zone is 07/13/2026, 16:11 GMT+5:30

Start time (GMT+5:30)
Schedule a specific date and time for the first scheduled action to run

Date
2026/07/13
Format: YYYY/MM/DD

Time
16:20:00
Format: hh:mm:ss

Recurrence
Once

Capacity adjustments
If the current capacity is below the minimum capacity at the scheduled time, capacity will scale out to the minimum capacity limit. If the current capacity is above the maximum capacity, it will scale in to the maximum capacity.

Current capacity limits
2-10

Specify adjusted capacity limits
Specify how the capacity limit should change when the scheduled action takes place.

Minimum number of tasks
4

Maximum number of tasks
10

Scheduled actions are usually used in pairs. Create one to scale-out and one to scale-in

[Cancel](#) [Create scheduled action](#)

Step 4 – Create the scale-in scheduled action

- Click **Create schedule action** again.
- Enter the scale-in policy name: SC-Weekend-Scale-In
- Set the same time zone and date.
- Choose a later trigger time for scale-in.

Create scheduled action Info

Scheduled actions add or remove capacity automatically at preset times or intervals. Amazon Elastic Container Service autoscaling is using the following Service-linked role: [arn:aws:iam::463646775279:role/aws-service-role/ecs.application-autoscaling.amazonaws.com/AWSServiceRoleForApplicationAutoScaling_ECSService](#)

Action details

Action name
SC-Weekend-Scale-In

Schedule
Specify a schedule for your capacity changes.

Time zone
Asia/Calcutta

Current time in selected time zone is 07/13/2026, 16:16 GMT+5:30

Start time (GMT+5:30)
Schedule a specific date and time for the first scheduled action to run

Date
2026/07/13
Format: YYYY/MM/DD

Time
16:25:00
Format: hh:mm:ss

Recurrence
Once

Capacity adjustments
If the current capacity is below the minimum capacity at the scheduled time, capacity will scale out to the minimum capacity limit. If the current capacity is above the maximum capacity, it will scale in to the maximum capacity.

- Set the desired count to return from 4 to 2.
- Set the Max count from 10 to 2 for proper verification.
- Save the scheduled scale-in action.

Start time (GMT+5:30)
Schedule a specific date and time for the first scheduled action to run

Date
2026/07/13
Format: YYYY/MM/DD

Time
16:25:00
Format: hh:mm:ss

Recurrence
Once

Capacity adjustments
If the current capacity is below the minimum capacity at the scheduled time, capacity will scale out to the minimum capacity limit. If the current capacity is above the maximum capacity, it will scale in to the maximum capacity.

Current capacity limits
2-10

Specify adjusted capacity limits
Specify how the group size should change when the scheduled action takes place.

Minimum number of tasks
2

Maximum number of tasks
2

Scheduled actions are usually used in pairs. Create one to scale-out and one to scale-in

[Cancel](#) [Create scheduled action](#)

Step 5 – Wait for the scheduled actions to run

- Keep the service running until the scale-out time is reached.
- Verify that the desired task count increases from 2 to 4.

Amazon Elastic Container Service

Clusters > my-fargate-cluster > Services > My-Web-App-Final > Tasks

My-Web-App-Final Info

Status: Active

Tasks (4 Desired): 2 running | 2 pending

Task definition: revision My-Web-App:1

Deployment status: Success

Health and metrics | **Tasks** | Logs | Deployments | Events | Configuration and networking | Service auto scaling | Event history | Tags

Tasks (1/4)

Filter tasks by property or value

Task	Last status	Desired status	Task definition	Health status	Started by	Created at
144e364d06ef45cb87e89...	Activating	Running	My-Web-App:1	Unknown	ecs-svc/20287024318...	33 seconds ago
4f080764cc1249d595e15...	Running	Running	My-Web-App:1	Unknown	ecs-svc/20287024318...	32 minutes ago

Containers for task 144e364d06ef45cb87e899c177371ee6

Containers (1)

Filter containers

- Wait until the scale-in time is reached.
- Verify that the desired task count decreases back to 2.

Amazon Elastic Container Service

Clusters > my-fargate-cluster > Services > My-Web-App-Final > Tasks

My-Web-App-Final Info

Status: Active

Tasks (2 Desired): 2 running | 0 pending

Task definition: revision My-Web-App:1

Deployment status: Success

Health and metrics | **Tasks** | Logs | Deployments | Events | Configuration and networking | Service auto scaling | Event history | Tags

Tasks (1/4)

Filter tasks by property or value

Task	Last status	Desired status	Task definition	Health status	Started by	Created at
144e364d06ef45cb87e89...	Running	Running	My-Web-App:1	Unknown	ecs-svc/20287024318...	5 minutes ago
c5a8bca4e8e44baf895aa...	Running	Running	My-Web-App:1	Unknown	ecs-svc/20287024318...	5 minutes ago
4f080764cc1249d595e15...	Deactivating	Stopped	My-Web-App:1	Unknown	ecs-svc/20287024318...	37 minutes ago
dd2d41f9d72b4b64b70d...	Deactivating	Stopped	My-Web-App:1	Unknown	ecs-svc/20287024318...	37 minutes ago

Containers for task 144e364d06ef45cb87e899c177371ee6

Containers (1)

Filter containers

Step 6 – Verify load balancer and target registration

- Open the application through the **Application Load Balancer** DNS name.
- Refresh the application and confirm that the container IDs change across tasks.
- Open the **EC2** target group view.
- Confirm that the newly created tasks are registered automatically as healthy targets.

The screenshot shows the AWS Management Console interface for the 'ecs-app-tg' target group. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The left sidebar shows the 'EC2' menu with options like Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, and Capacity Manager. The main content area displays the 'Targets' tab for the 'ecs-app-tg' target group. It shows a summary of 4 total targets, 4 healthy, 0 unhealthy, 0 unused, 0 initial, and 0 draining. Below this, there's a section for 'Distribution of targets by Availability Zone (AZ)' and a table of 'Registered targets (4)'. The table has columns for IP address, Port, Zone, Health status, Health status details, Administrative override, Overri..., and Anomaly detectio... The first four targets are highlighted with a red box.

IP address	Port	Zone	Health status	Health status details	Administrative override	Overri...	Anomaly detectio...
192.168.0.224	80	us-east-1b (...)	Healthy	-	No override	No overri...	Normal
192.168.0.171	80	us-east-1a (...)	Healthy	-	No override	No overri...	Normal
192.168.0.183	80	us-east-1a (...)	Healthy	-	No override	No overri...	Normal
192.168.0.196	80	us-east-1b (...)	Healthy	-	No override	No overri...	Normal

Verification

- Verified that **Amazon ECS Service Auto Scaling** was enabled.
- Verified that the minimum and maximum task limits were set correctly.
- Verified that the scale-out scheduled action increased the service from 2 tasks to 4 tasks.
- Verified that the scale-in scheduled action reduced the service from 4 tasks back to 2 tasks.
- Verified that the **Application Load Balancer** kept routing traffic during scaling.
- Verified that the ECS tasks were automatically registered in the target group.
- Verified that scheduled scaling worked correctly for both scale-out and scale-in actions.